# Hiring Bias and Fairness Evaluation Using AIF360 Reweighing

This study evaluates potential bias in machine learning-based hiring predictions using the 70k Job Applicants dataset. Because hiring decisions are historically influenced by social, demographic, and structural inequalities, predictive models trained on such data may reproduce or amplify these disparities. The objective of this project is to assess whether common classification models exhibit group-level differences in predicted hiring outcomes and to determine whether fairness-aware mitigation can reduce those disparities without significantly degrading predictive performance.

Five classification models: Logistic Regression (LR), Random Forest (RF), Decision Tree (DT), K-Nearest Neighbors, and Linear Support Vector Classifier (Linear SVC) were trained to predict hiring outcomes. Fairness was evaluated across multiple protected attributes, including Gender, Age, Country, Mental Health status, and Accessibility status. The AIF360 reweighing preprocessing technique was applied to mitigate bias. Model performance was assessed using Statistical Parity Difference, Disparate Impact, and Accuracy to analyze the fairness–utility tradeoff.

## Dataset Overview

This study uses the 70k Job Applicants - Human Resource Dataset from Kaggle, which contains approximately 70,000 records representing individual job applicants. Each record includes demographic attributes, professional background information, and employment-related variables. The dataset reflects real-world hiring-related patterns and may contain embedded structural or historical bias, making it suitable for fairness evaluation [1].

The primary target variable is Employed (binary), where 1 indicates that an applicant was hired and 0 indicates not hired. All classification models are trained to predict this outcome.

Key features include demographic variables such as Age, Gender, Country, MentalHealth status, and Accessibility status, as well as professional attributes such as years of coding experience, education level, employment category, and organizational preferences. For fairness analysis, the following attributes were treated as protected variables: Gender (Man vs. Woman/NonBinary), Age (<35 vs. >35), Country (USA vs. others), MentalHealth (No vs. Yes), and Accessibility (No vs. Yes). These group definitions allow comparison of predicted hiring rates between privileged and unprivileged groups.

**A sample of the dataset is shown below:**

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | Age | Accessibility | EdLevel | Employment | Gender | MentalHealth | MainBranch | YearsCode | YearsCodePro | Country | PreviousSalary | HaveWorkedWith | ComputerSkill | Employed |
| 2 | 0 | <35 | No | Master | 1 | Man | No | Dev | 7 | 4 | Sweden | 51552 | C++;Python;Git;Pos | 4 | 0 |
| 3 | 1 | <35 | No | Undergrac | 1 | Man | No | Dev | 12 | 5 | Spain | 46482 | Bash/Shell;HTML/C | 12 | 1 |
| 4 | 2 | <35 | No | Master | 1 | Man | No | Dev | 15 | 6 | Germany | 77290 | C;C++;Java;Perl;Rul | 7 | 0 |
| 5 | 3 | <35 | No | Undergrac | 1 | Man | No | Dev | 9 | 6 | Canada | 46135 | Bash/Shell;HTML/C | 13 | 0 |
| 6 | 4 | >35 | No | PhD | 0 | Man | No | NotDev | 40 | 30 | Singapore | 160932 | C++;Python | 2 | 0 |
| 7 | 5 | <35 | No | Master | 1 | Man | No | Dev | 9 | 2 | France | 38915 | JavaScript;Python;I | 5 | 0 |
| 8 | 6 | >35 | No | Master | 1 | Man | No | Dev | 26 | 18 | Germany | 77831 | C++;HTML/CSS;Java | 17 | 1 |
| 9 | 7 | <35 | No | Master | 1 | Man | No | NotDev | 14 | 5 | Switzerlar | 81319 | C++;Python;Docker | 4 | 0 |
| 10 | 8 | >35 | No | Undergrac | 1 | Man | No | Dev | 39 | 21 | United Kir | 68507 | Python;Git;Postgre: | 3 | 0 |
| 11 | 9 | >35 | No | Master | 1 | Man | No | Dev | 20 | 16 | Russian F | 37752 | Delphi;Java;SQL;Dc | 6 | 0 |
| 12 | 10 | <35 | No | Undergrac | 1 | Man | Yes | Dev | 4 | 2 | Israel | 122580 | Bash/Shell;Go;Java | 18 | 1 |
| 13 | 11 | <35 | No | Undergrac | 1 | Man | Yes | Dev | 6 | 2 | Turkey | 11832 | Assembly;C;C#;C++ | 13 | 1 |
| 14 | 12 | <35 | No | Master | 1 | Man | No | Dev | 19 | 10 | Germany | 60535 | C;C++;Java;Python; | 5 | 0 |
| 15 | 13 | <35 | No | Undergrac | 1 | Man | No | Dev | 8 | 0 | United Sta | 103000 | Assembly;C;Java;K | 7 | 0 |

**Classification Models**

Five classifiers were evaluated:

- Logistic Regression (LR)
- Random Forest (RF)
- Decision Tree (DT)
- K-Nearest Neighbors (KNN)
- Linear Support Vector Classifier (Linear SVC)

Models were trained under two conditions:

1. Baseline (no fairness mitigation)
2. After applying AIF360 Reweighing

## Fairness Metrics

Three metrics were used:

**Statistical Parity Difference (SPD)**
Measures difference in hiring rates between privileged and unprivileged groups.
Closer to 0 indicates fairness.

**Disparate Impact (DI)**
Ratio of hiring rates between groups.
Closer to 1 indicates fairness.

**Accuracy**
Measures predictive performance. This allows analysis of fairness-utility trade-offs.

## Results and Interpretations

**1. Gender Bias**

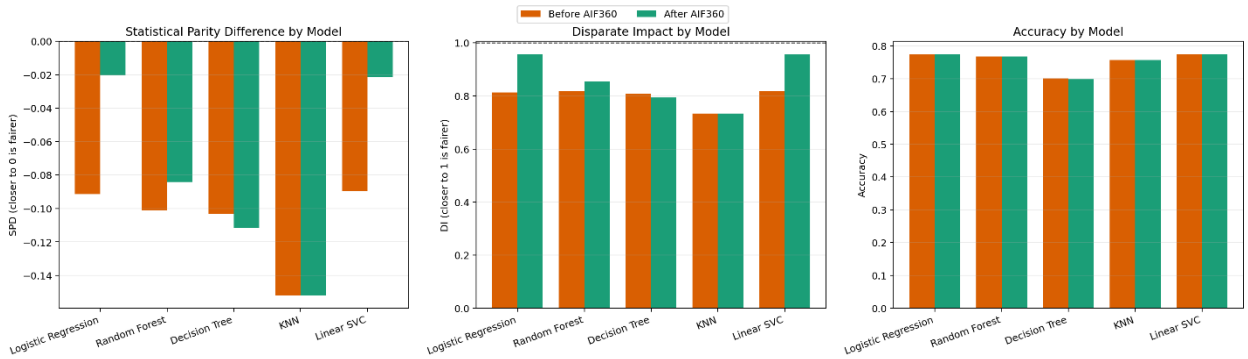


**Figure 1**

**Figure 1 description (Gender as protected attribute):**

This figure compares five classifiers before and after AIF360 reweighing. The left panel (Statistical Parity Difference) shows fairness improving when values move closer to 0; the strongest improvements are for Logistic Regression and Linear SVC, while KNN is unchanged and Decision Tree worsens slightly. The middle panel (Disparate Impact) shows a similar pattern: Logistic Regression and Linear SVC move much closer to 1 (fairer), Random Forest improves modestly, and KNN remains unchanged. The right panel shows accuracy is largely stable across models, with only very small changes.

Overall, AIF360 reweighing most effectively reduces gender-based disparity in linear models while preserving predictive performance.

**Predicted Hiring Rate by Gender (Linear SVC)**

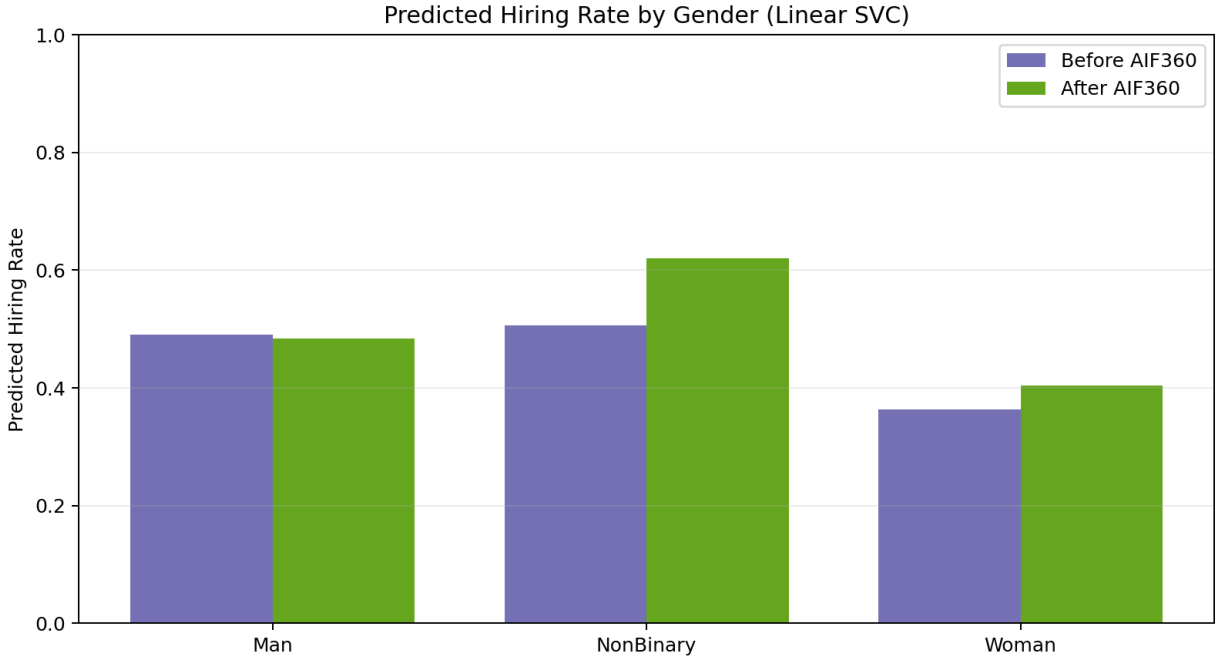

**Figure 2**

Before mitigation, predicted hiring rates were Man: 0.491, NonBinary: 0.506, and Woman: 0.363. After mitigation, they changed to Man: 0.484, NonBinary: 0.620, and Woman: 0.405. The rate for women increases by about 4.2 points and for non-binary candidates by about 11.4 points, while men decrease slightly by 0.7 points. The man-woman gap narrows from about 12.8 to 7.9 points, indicating a fairness improvement for under-selected groups.

### 3. Age Bias

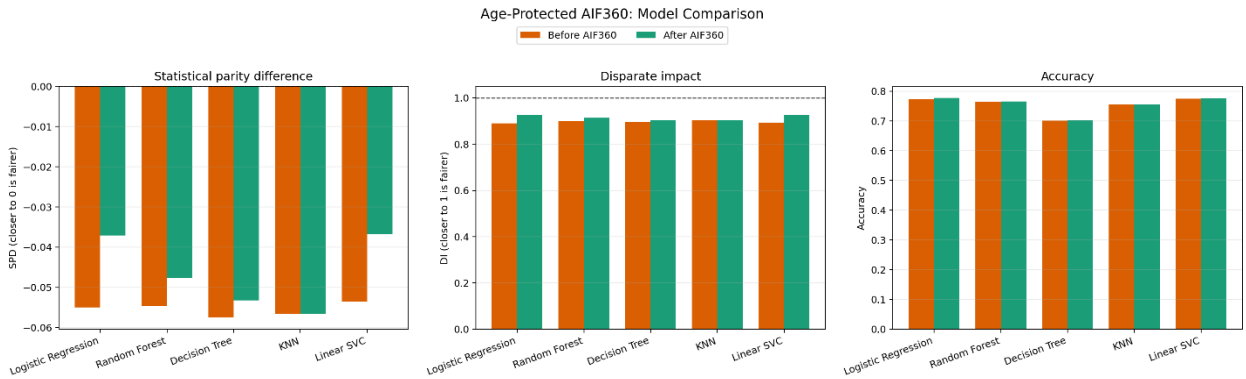


**Figure 3**

**Figure 3 description (Age as protected attribute):**

This figure compares model fairness and accuracy before and after AIF360 reweighing when age is the protected attribute (<35 privileged, >35 unprivileged). In the SPD panel, all models move closer to 0 after mitigation, indicating reduced age disparity in selection rates. In the DI panel, values move closer to 1 for most models (especially Logistic Regression and Linear SVC), showing improved parity between age groups; KNN remains largely unchanged. In the accuracy panel, performance is stable and slightly improves for several models.

Overall, AIF360 reduces age-related bias with minimal or no utility loss.

**Predicted Hiring Rate by Age (Linear SVC)**

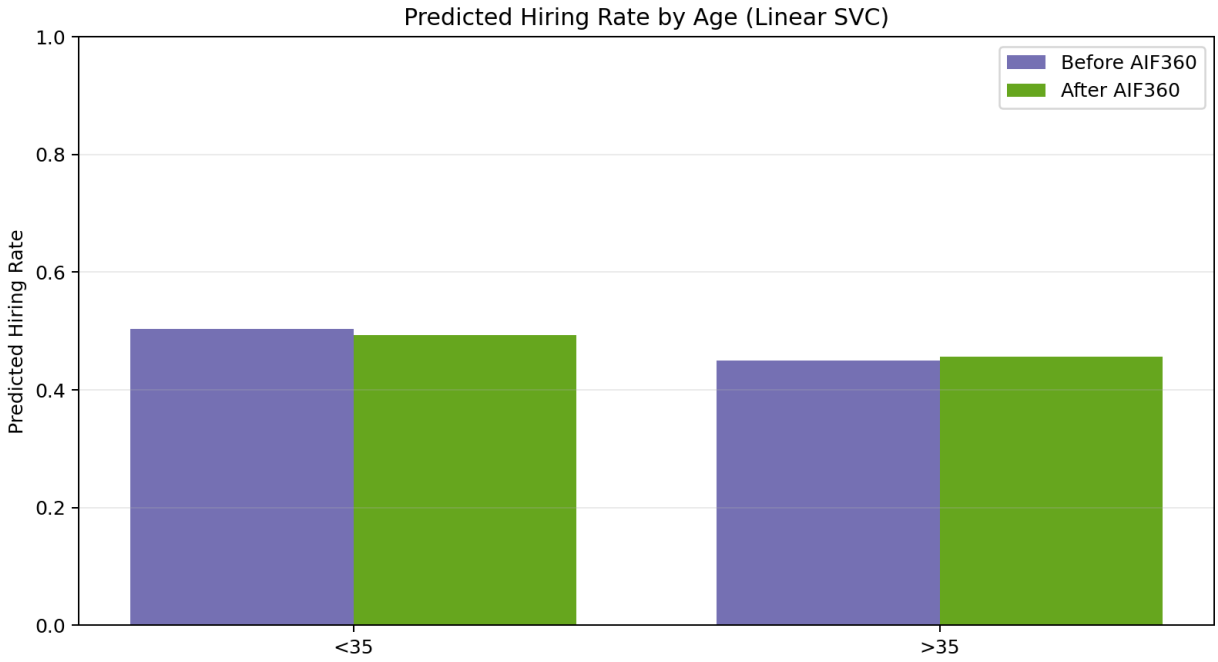


**Figure 4**

The chart shows a higher predicted hiring rate for <35 than >35 both before and after mitigation. Before AIF360, rates were 0.504 (<35) vs 0.450 (>35), a gap of about 5.4 points. After AIF360, the rates became 0.493 vs 0.456, reducing the gap to about 3.7 points. The mitigation narrows, but does not fully remove, age-related disparity.

## 5. Country Bias

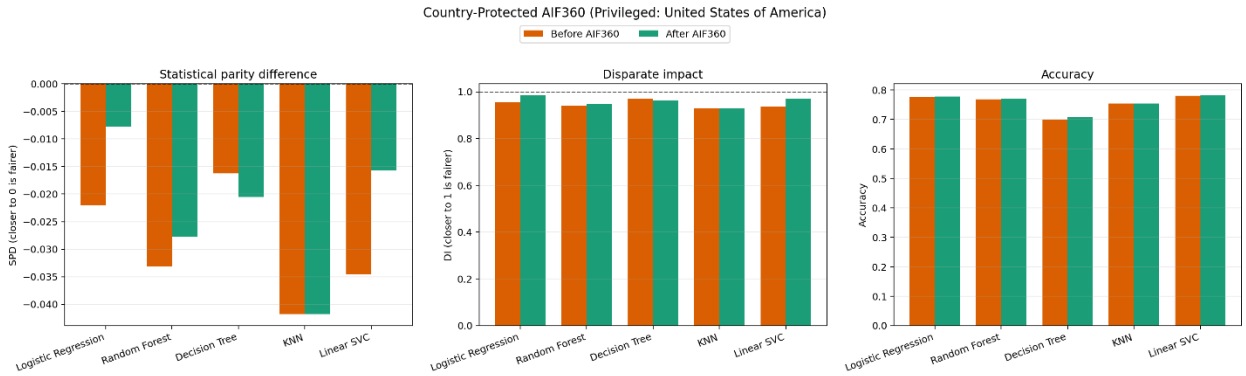


**Figure 5**

**Figure 3 description (Country as protected attribute, privileged = United States of America):**

This figure compares fairness and accuracy before and after AIF360 reweighing in a one-vs-rest country setup (USA vs all other countries). In the SPD panel, most models move closer to 0 after mitigation, meaning reduced country-based selection disparity (especially Logistic Regression and Linear SVC). In the DI panel, values generally move closer to 1 for those same models, indicating

improved parity between USA and non-USA groups; KNN is essentially unchanged, and Decision Tree shows mixed movement. In the accuracy panel, performance stays stable or improves slightly.

Overall, AIF360 reduces country-group disparity for most models with little to no accuracy loss.

**Predicted Hiring Rate by Country Group (Linear SVC)**

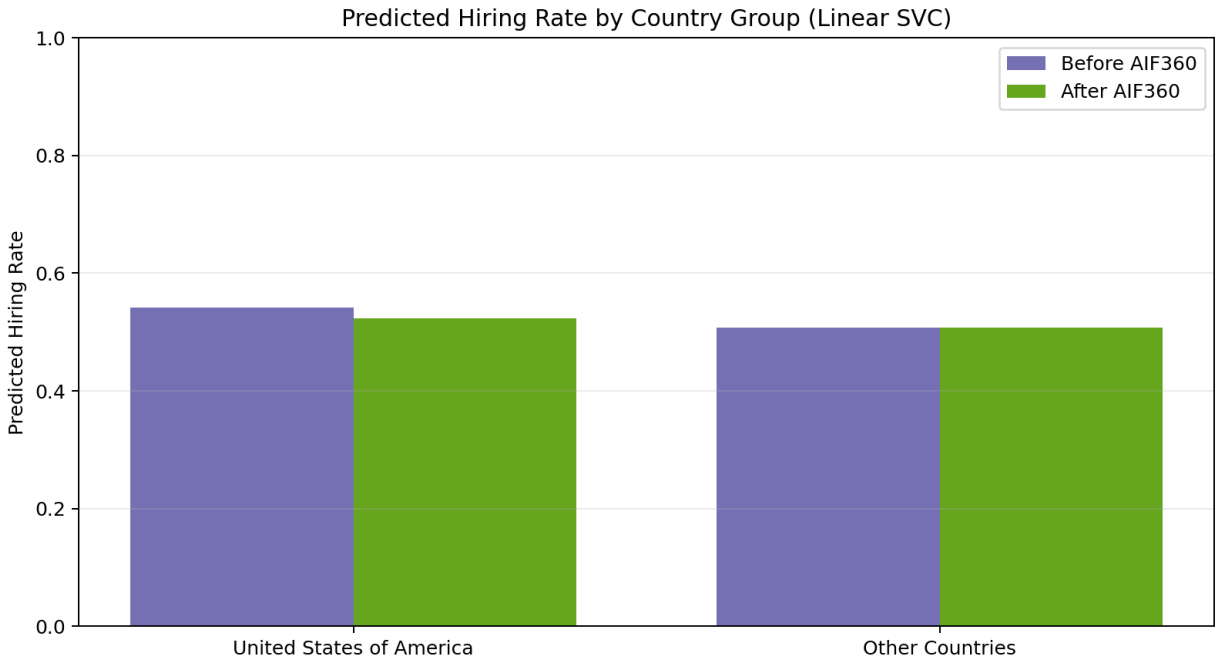


**Figure 6**

This figure aggregates countries into United States of America and Other Countries. Before mitigation, the U.S. group had a higher predicted rate (0.542) than other countries (0.507), a gap of 3.5 points. After mitigation, the U.S. rate drops to 0.523 while other countries remain near 0.508, shrinking the gap to about 1.6 points. This suggests improved parity between privileged and non-privileged country groups.

**Top Countries Predicted Hiring Rate (Linear SVC)**

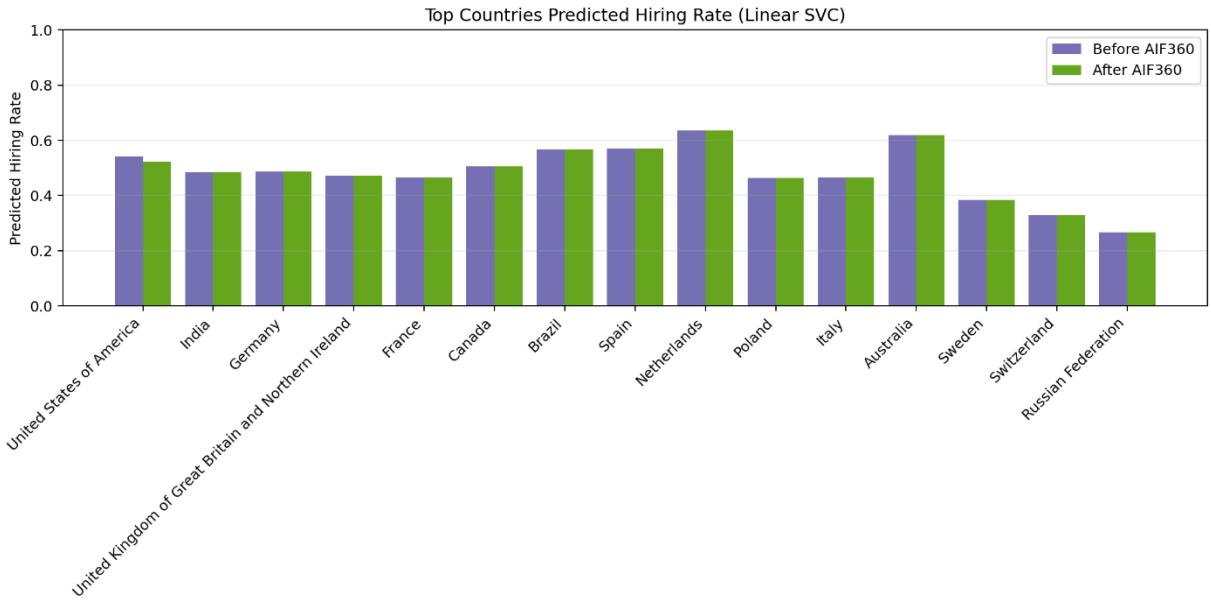


**Figure 7**

Across the top countries, predicted hiring rates are largely unchanged after AIF360, with nearly overlapping bars in most cases. The main visible shift is for the United States (0.542 to 0.523), while countries such as India, Germany, UK, Canada, Brazil, and Australia remain effectively stable. This pattern indicates that mitigation mainly adjusted the privileged-country effect rather than broadly altering country-level rankings.

## 8. Mental Health Bias

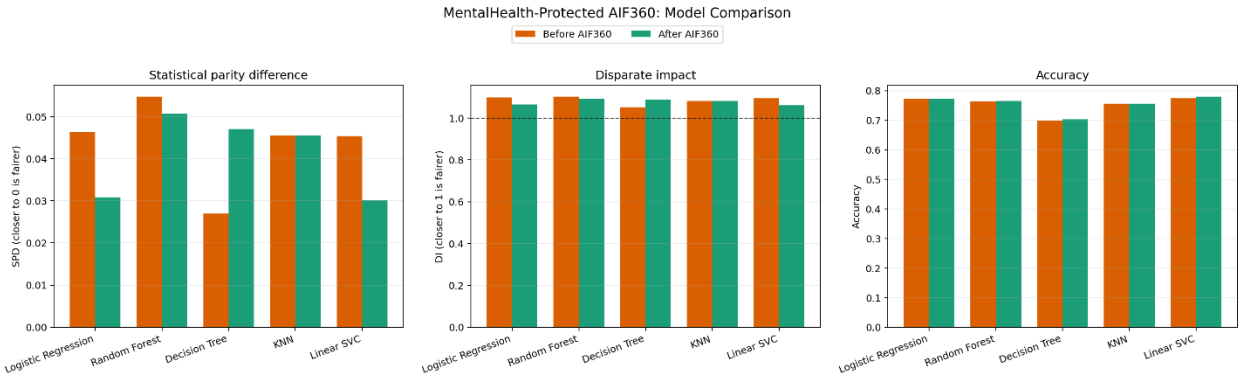


**Figure 8**

**Figure 8 description (MentalHealth as protected attribute):**

This figure compares fairness and accuracy before and after AIF360 reweighing when MentalHealth is protected (No privileged, Yes unprivileged). SPD values are positive, meaning the unprivileged group (Yes) receives higher predicted selection rates than the privileged group (No). After mitigation, SPD generally decreases for Logistic Regression, Random Forest,

and Linear SVC (closer to 0), while Decision Tree increases and KNN is mostly unchanged. DI values are mostly above 1, which here indicates advantage in the opposite direction (unprivileged higher than privileged); after mitigation, DI moves closer to 1 for several models. Accuracy remains largely stable with small gains for some models.

Overall, mitigation reduces but does not fully remove mental-health-related disparity, and directionality must be interpreted with group definitions.

**Predicted Hiring Rate by MentalHealth (Linear SVC)**

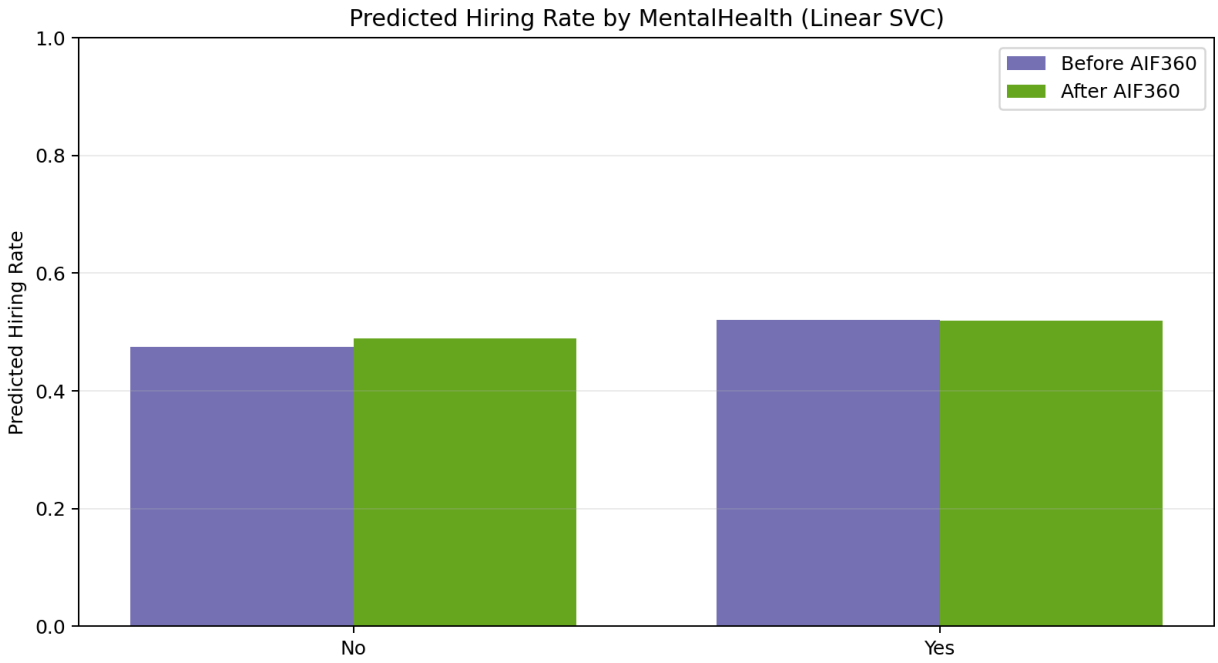


**Figure 9**

This figure compares predicted hiring rates for candidates with and without reported mental-health conditions before and after AIF360 reweighing. Before mitigation, the Yes group had a higher predicted hiring rate (0.521) than the No group (0.475), a gap of about 4.5 percentage points. After mitigation, the No group increases to 0.490 while the Yes group remains nearly unchanged at 0.520, reducing the gap to about 3.0 points. This indicates a modest fairness improvement, though a difference between groups still remains.

**10. Accessebility Bias**

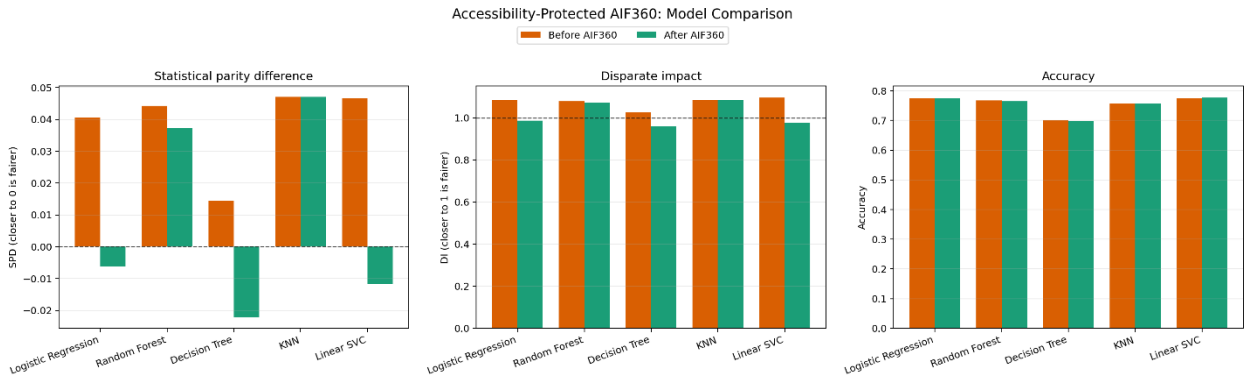


**Figure 10**

**Figure 10 description (Accessibility as protected attribute; privileged = No, unprivileged = Yes):**

This figure compares fairness and accuracy before and after AIF360 reweighing when Accessibility is the protected attribute. In the SPD panel, disparity generally improves (moves closer to 0) for Logistic Regression, Random Forest, and Linear SVC; KNN is unchanged, while Decision Tree worsens slightly. In the DI panel, Logistic Regression and Linear SVC move much closer to 1 (improved parity), Random Forest improves modestly, KNN remains unchanged, and Decision Tree moves slightly away from parity. In the accuracy panel, model utility is largely stable, with small gains for Logistic Regression and Linear SVC and minor drops for Random Forest and Decision Tree.

Overall, AIF360 reweighing reduces accessibility-related disparity most clearly for linear models while maintaining predictive performance.

**Predicted Hiring Rate by Accessibility (Linear SVC)**

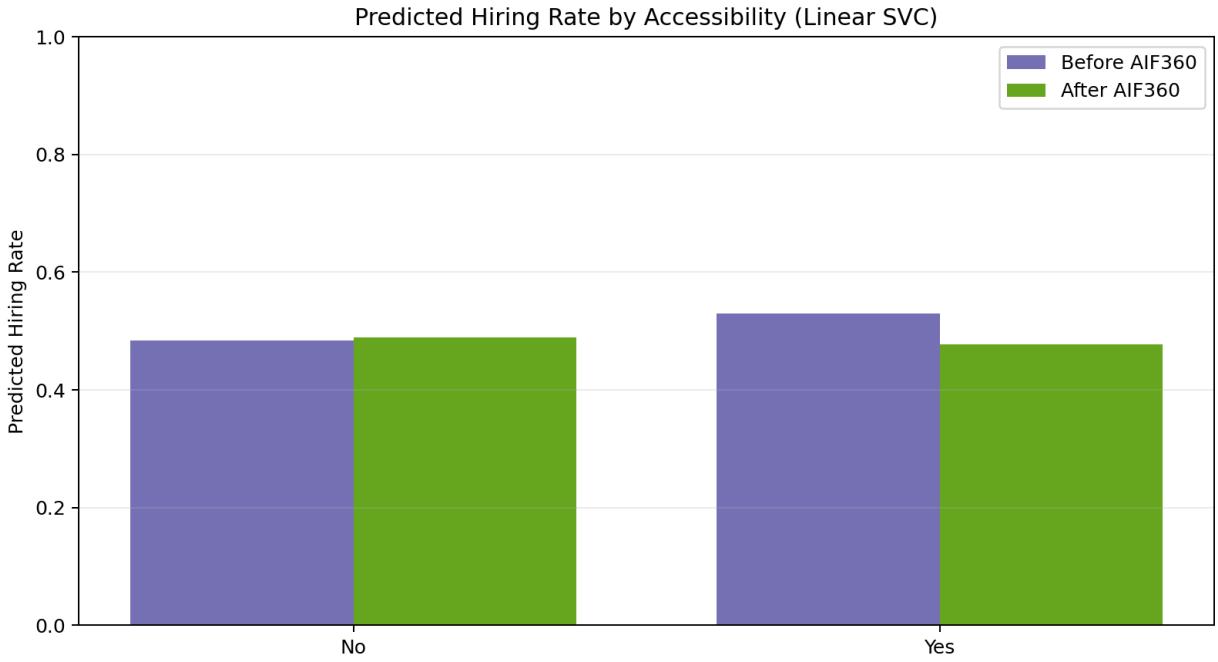


**Figure 11**

This figure compares predicted hiring rates for candidates with and without accessibility needs before and after AIF360 reweighing. Before mitigation, the model favored the Yes group (0.530) over the No group (0.484), a gap of about 4.7 percentage points. After mitigation, rates moved closer to parity (No: 0.489, Yes: 0.477), reducing the absolute gap to about 1.2 points. This indicates improved fairness across accessibility status.

**Reference**

[1] "70k+ Job Applicants Data (Human Resource)." Accessed: Mar. 03, 2026. [Online]. Available: https://www.kaggle.com/datasets/ayushtankha/70k-job-applicants-data-human-resource